

1. Architectural Style Analysis

1.1 Layered Architecture

Structure:

The system is divided into different layers such as Presentation, Business Logic, and Data layers.

Advantages:

- Provides clear separation of concerns.
- Easy to understand and maintain.
- Changes in one layer can be isolated from other layers.
- Suitable for applications with a straightforward processing flow.

Disadvantages:

- Communication through multiple layers can introduce performance overhead.
- Individual functions cannot be easily scaled independently.
- Changes or failures in one layer can affect dependent layers.
- Can result in tighter coupling between layers.

Suitability for the Warehouse Inventory & Pallet Location Tracker:

Layered Architecture can organize the warehouse application clearly, but independently scaling functions such as inventory lookup and rack monitoring would be more difficult.

1.2 Microservices Architecture

Structure:

The system is divided into small, independent services, with each service responsible for a specific function.

Advantages:

- Services can be developed and deployed independently.
- Individual services can be scaled independently.
- Failure in one service can be isolated from other services.
- Different services can use technologies suited to their individual requirements.

Disadvantages:

- More complex to develop and manage.
- Communication between services introduces network latency.
- Maintaining consistency between services can be challenging.
- Requires monitoring and management of multiple services.

Suitability for the Warehouse Inventory & Pallet Location Tracker:

Microservices Architecture fits the warehouse system because its functions can be separated into services such as Pallet Management, Location Tracking, Placement Validation, Rack Capacity Monitoring, and Inventory Lookup.

1.3 Client-Server Architecture

Structure:

Multiple clients communicate with a centralized server that handles processing and data management.

Advantages:

- Provides centralized control of warehouse data.
- Makes data management easier.
- Multiple warehouse clients can share the same server.
- Relatively simple to deploy.

Disadvantages:

- The central server can become a bottleneck.
- Server failure can affect multiple warehouse operations.
- Individual functions cannot be scaled independently.
- High-volume operations such as searching across 100,000 bins can place significant load on the server.

Suitability for the Warehouse Inventory & Pallet Location Tracker:

Client-Server Architecture provides centralized control, but the centralized server can limit scalability and fault isolation for the warehouse system.

2. Architecture Selection and Justification

Architectural Choice

I chose Microservices Architecture for the Warehouse Inventory & Pallet Location Tracker.

Reason 1 – Independent Functionality

The warehouse system contains several distinct functions, including pallet identification, placement validation, location tracking, rack capacity monitoring, and inventory lookup. These functions can be implemented as separate services, allowing each service to perform a specific responsibility independently.

Reason 2 – Independent Scalability

Different warehouse functions may experience different workloads. For example, the Inventory Lookup Service may receive a large number of search requests, while rack capacity monitoring may have a lower workload. Microservices Architecture allows individual services to be scaled independently based on their workload instead of scaling the entire system.

Security Advantage

Microservices Architecture provides service-level isolation. Access to sensitive operations can be restricted so that each service only exposes the interfaces required by other services. For example, the Placement Validation Service can access rack capacity information through a defined interface without directly accessing the entire warehouse data system.

Performance Benefit

The Inventory Lookup Service can be optimized and scaled independently to support the requirement of locating an SKU pallet coordinate across 100,000 bins in under 100 ms. This allows high-volume lookup operations to be handled efficiently without unnecessarily affecting other warehouse functions.

Conclusion

I selected Microservices Architecture because its independent services match the warehouse system's separate functional requirements while supporting independent scaling, service isolation, and performance optimization.